

Everything You Wanted to Know About Mobile Accessibility

Making applications accessible to all categories of users is an important design principle to keep in mind when developing successful, universal applications. Accessibility challenges posed by mobile devices are often due to the presence of specialised hardware sensors, interface mechanisms, comparatively smaller-sized displays, etc. This article provides insights into the various aspects of mobile accessibility, covering the myths about accessibility, accessibility dimensions and the steps involved in making mobile applications truly accessible.

One of the important factors in designing a tool or application is *accessibility (A11Y)*. Mobile accessibility refers to the process of making mobile apps usable for all groups of users, which includes persons with disabilities and the elderly. A United Nations survey has estimated that there are one billion people with disabilities in the world, which amounts to approximately 15 per cent of the global population. The wish list of a developer should include the task of making the application accessible to all categories of users. The starting point in designing universally accessible applications is to understand the fact that accessibility is not an add-on layer but is a process that covers the multiple dimensions of application design.

Myths related to accessibility

Some of the myths about the critical barriers preventing applications from being accessible are listed below:

- Incorporating accessibility in applications or mobile apps will make them look dull and unattractive.
- Accessibility will not make a considerable impact on the reach of the mobile app.
- Building accessible apps will prolong the mobile application development process unreasonably.
- Making the mobile apps accessible requires highly complicated skills.

All the above mentioned myths can be addressed by educating developers and designers regarding the easy-to-follow steps in the accessibility process. In reality, accessibility would make the applications look more professional and world class. The presence of a sizeable population with disabilities and in the elderly age group certainly is a motivation to make an application



accessible. Moreover, accessibility options will be adopted by other users as well, in specific scenarios. For example, an application with a voice interface is useful while driving. The standard accessibility practices can be managed well within a reasonable time frame.

WCAG 2.0

The Web Content Accessibility Guidelines (WCAG) facilitate accessibility of Web content. WCAG covers four major dimensions (POUR) that ensure an app is accessible:

- Perceivable (P)
- Operable (O)
- Understandable (U)
- Robust (R)

Perceivable features

With regards to perceivable features, three major factors are considered with respect to mobile app accessibility.

Small screen size: Keeping control on the quantity of information presented to a user in a single page needs to be given priority. Provision of separate accessible style sheets also serves as an effective mechanism. Maintaining reasonable sizes for touch controls in order to minimise the need to zoom should be the norm to help persons with poor vision.

Zoom/Magnification: The zoom levels can be managed either at the operating system level or at the browser level. The magnification settings, which apply to the entire screen and across applications, are controlled through *Accessibility Settings*. The browser level settings enable the controlling of text rendered in the view port of the browser.

Contrast: As the mobile devices are used across various environments with different lighting conditions, the contrast level plays an important role. The WCAG 2.0 regulations indicate a minimum contrast of 4.5:1.